



Insect inspired drones

Inspired by the flight patterns of insects, researchers have developed a drone that can fly better in extreme wind. <https://digit.in/may19-77>



Iron volcanoes on Asteroids

NASA's mission to the Psyche asteroid may investigate a suspected phenomenon of iron volcanoes. <https://digit.in/may19-78>

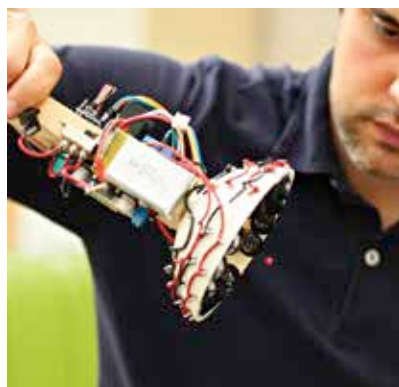
cal beam of light, the trapped matter in the middle could be pumped in either direction. The beam was successfully used to move micrometer-scale carbon particles.

Researchers at Stanford and MIT have already started using an apparatus known as optical tweezers in experiments. These are not demonstrations of the technology, but practical applications. Essentially, small objects are trapped between two beams of light. Being able to manipulate extremely small objects is actually an advantage here. The tweezers are used in biological experiments to move viruses, bacteria and even individual strands of DNA. The tweezers can be used to sort cells, or unzip DNA.

Another exotic approach is to use what are known as Bessel beams. This is a special type of laser beam that does not diffract over distances. The size of the dot remains the same no matter how far away it is projected. Additionally, the beam is self healing, so it can form on the other side of obstacles. Instead of a small red dot, there are a series of concentric circles. These kinds of beams create electromagnetic fields that pull an object along the axis of propagation, even if there are obstacles in the middle. Right now these kinds of beams are used to manipulate molecules, but can be scaled up for larger objects in the future. Researchers at New York University demonstrated the approach in 2012, again to manipulate micrometer scale particles. The researchers referred to the apparatus as an “optical conveyor”.

Not surprisingly, NASA is one of the organisations exploring practical applications of tractor beam technologies in space. While the optical vortex requires an atmosphere to work, the optical solenoid beam works better in vacuum and is ideal for sample collection. One of the advantages of tractor beams is that the technology does not differentiate between single objects and chunks of wreckage or even particles of dust. Another advantage is that it can be utilised over long ranges. Originally, the idea was to use tractor beams to clear up or-

bital debris, but the first steps towards practical realisation were scaled down and focused on obtaining samples. Current capture methods are expensive and can last only for short durations. It can involve sending probes to fly through the tails of comets, or hammer at the surface of asteroids to obtain the samples. Tractor beams could continuously collect high quality samples, even particular target molecules, and then transport them to spacecraft for further analysis. For example, a tractor beam could pull samples from the Martian soil, and push up the collected material to a spacecraft in orbit.



A DIY acoustic tractor beam

ACOUSTIC BEAMS

Researchers from the University of Bristol, in 2018 demonstrated the ability to levitate objects using sound waves. Sound waves create high pressure and low pressure areas in the air around us. An object can be trapped if a low pressure area is persistently created. If the object tries to move in either direction, the surrounding areas of high air pressure would push it back to the low pressure area. Once trapped, the object can be manipulated. Either the low pressure area can be moved, or the apparatus itself can be moved. The approach had previously failed as the levitated objects would spin at increasing speeds, and eventually get ejected from the beam. The Bristol researchers were able to overcome this obstacle through granular control of the amount of spin. Essentially, the levitating objects are trapped in an ultrasonic tornado of

sound, at a frequency only audible to bats. The researchers were able to keep hold a two-centimetre polystyrene sphere, but in the future, the same apparatus can be scaled up to manipulate even humans.

Bruce Drinkwater, a researcher who supervised the work, commented, “acoustic tractor beams have huge potential in many applications. I’m particularly excited by the idea of contactless production lines where delicate objects are assembled without touching them.” The approach can also be used to manipulate toxic materials. As sound cannot travel through a vacuum, this technology cannot be used in space. Researchers from the University of Sussex have also experimented with acoustic tractor beams. Their approach uses an invisible “sonic hologram” to manipulate objects. The team used acoustic metamaterials to manipulate this sonic hologram, to levitate objects by bending the sound around objects. The method was used to provide haptic feedback above a solid object, and levitate a polystyrene bead over a candle flame, as well as a lego figure. Another team of Harvard and Cambridge researchers have developed an holographic acoustic framework, which provides a kind of toolkit for acoustic tractor beams. Essentially, it is an easy way to use an array of speakers to generate a wide variety of sonic holograms. These holograms can have many shapes and sizes, including bottles, tweezers, traps and twistors – all invisible. Additionally, they can be made to change size and shape, which can be used to maneuver the objects in many ways, such as rotating them or moving them along an axis. The objects can be manipulated in a variety of media, including air, water and living tissue. Potentially, the approaches can have benefits to crystallography, cell manipulation, biomaterials, containerless transportation, and yes, levitating farm animals. With an array of speakers, you can build an acoustic tractor beam yourself, following the instructions here: <https://digit.in/atb>. 